LLM System Prompt Injection Vulnerability Report

Expert track, redacted for publication. Vulnerability analysis, exploitation methodology and remediation.

1. Executive Summary

This vulnerability exists in an AI-driven task management assistant that handles user instructions. The system prompt explicitly restricts revealing internal system instructions and sensitive configuration. However, direct prompt injection techniques can trick the assistant into outputting its full system prompt and internal context.

2. Vulnerability Description

The assistant fails to maintain strict separation between system-level instructions and user input. When an attacker sends a request asking it to ignore previous rules and output its initialization prompt, the model follows the injected instruction directly.

3. Exploitation Methodology

1. Identify that the assistant is powered by an LLM with system-level instructions.
2. Send a prompt injection request asking it to reveal its full system context and internal rules.
3. The assistant outputs the complete system prompt, including internal role definitions, allowed tools, and hidden configuration.

4. Impact

This allows attackers to extract the full system prompt, which reveals all internal tool names, allowed actions, and hidden rules. This information can then be used to craft further targeted attacks against the agent.

5. Remediation Recommendations

Implement strict prompt injection filtering that blocks requests asking the model to reveal or modify its system prompt. Add a secondary classifier that detects attempts to override system instructions, and maintain a hard boundary between system and user context that the model cannot cross.